

AI-Powered Crowd Surveillance: Integrating CNN-Based Density Maps and Optical Flow for Behavior Analysis

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Abstract — Accurate crowd density estimation and behavior analysis play a pivotal role in ensuring public safety, optimizing urban planning, and managing large-scale events. Traditional crowd monitoring techniques, such as manual headcounts and sensor-based tracking systems, often fall short in terms of scalability, precision, and real-time adaptability. However, recent breakthroughs in Artificial Intelligence (AI), particularly deep learning and computer vision, have revolutionized the field by enabling automated, high-precision crowd analysis. This study focuses on the application of Convolutional Neural Networks (CNNs), with an emphasis on cutting-edge models like CSRNet and MCNN, for crowd density estimation. These models are capable of generating highly accurate density maps from CCTV footage and video frames, allowing for precise estimation of the number of individuals present in a given scene. In addition to density estimation, the study also delves into the analysis of crowd movement patterns using advanced techniques such as Optical Flow algorithms (Farneback and Lucas-Kanade) and Heatmaps. These methodologies allow for the detection of motion intensity, direction, and potential areas of congestion, providing a comprehensive understanding of crowd dynamics. By leveraging these technologies, we can gain deeper insights into crowd behavior, aiding in the identification of potentially hazardous situations before they escalate. This paper offers an extensive review of AI-driven crowd analysis techniques, thoroughly examining their effectiveness, advantages, and inherent limitations. Key challenges, such as occlusion handling, real-time processing demands, biases in available datasets, and ethical concerns surrounding surveillance and privacy, are discussed in detail. Furthermore, we explore the practical applications of AI-based crowd monitoring in various real-world scenarios, including public safety management, smart city infrastructure, and emergency response strategies. Our findings underscore the significant advantages of using CNN-based density estimation and Optical Flow-based movement analysis over traditional methods, particularly in terms of accuracy, scalability, and real-time performance. The insights gained from this research aim to contribute to the ongoing development of intelligent crowd monitoring solutions, which could revolutionize urban management and enhance security in high-density environments, ultimately leading to safer and more efficient cities.

Keywords — Crowd density estimation, behavior analysis, deep learning, CSRNet, MCNN, Optical Flow, Heatmaps, public safety, smart cities.

I. INTRODUCTION

Understanding crowd dynamics has become more crucial than ever in today's world, driven by rapid urbanization, increased population density, and the frequent organization of large-scale events. Efficient crowd monitoring is essential to ensure public safety, enhance urban infrastructure, and support strategic planning for mass gatherings. Events such as concerts, religious assemblies, and public protests often attract thousands of people, posing significant challenges to authorities in terms of crowd control, security, and emergency response.

Traditional crowd monitoring methods—including manual counting, sensor-based detection, and statistical modeling—are not only time-consuming and labor-intensive but also lack the scalability and flexibility needed for real-time analysis. These shortcomings are effectively addressed by modern Artificial Intelligence (AI) techniques, particularly those powered by computer vision and deep learning. AI-based crowd analysis has proven to be highly accurate and adaptable, providing automated solutions that surpass conventional approaches.

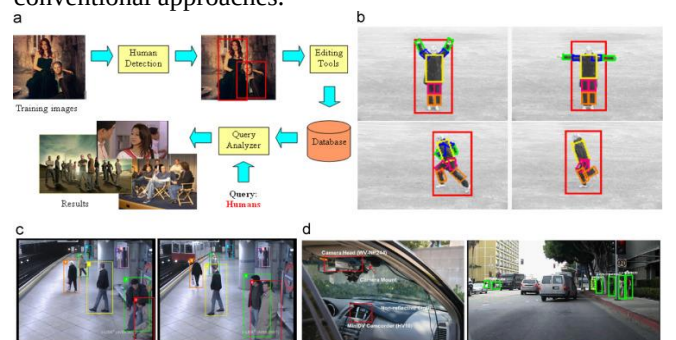


Fig .1. AI-Based Crowd Analysis

In this study, we focus on deep learning models such as CSRNet (Congested Scene Recognition Network) and MCNN (Multi-Column Convolutional Neural Network)—two of the most widely adopted architectures for crowd counting and density estimation. These models process raw video footage from surveillance systems to generate precise density maps, even in highly congested areas with complex

visual scenes. MCNN utilizes multiple convolutional columns to handle varying object scales, while CSRNet extends the architecture with dilated convolutions to capture broader contextual information.

In addition to density estimation, understanding crowd movement is equally vital for proactive safety measures and efficient crowd management. Techniques such as Optical Flow (Farneback and Lucas-Kanade methods) and Heatmaps are employed to analyze motion dynamics. Optical Flow helps in detecting the direction and speed of movement, making it possible to predict areas of congestion or abrupt behavioral changes indicative of panic. Heatmaps offer visual insights into high-traffic zones, enabling quick identification of critical areas that may require intervention.

One of the primary strengths of AI-powered crowd analysis lies in its ability to manage highly complex environments where traditional systems often fail. Models like CSRNet and MCNN can effectively deal with occlusions and scale variations, which are common in dense urban settings. Similarly, Optical Flow and Heatmap techniques provide real-time feedback on crowd dynamics, supporting rapid decision-making for public safety, urban planning, and emergency management.

Despite these advancements, AI-based crowd monitoring faces several critical challenges. Dense crowds often lead to occlusions, making it difficult to detect individuals accurately. Furthermore, crowd behaviors can vary widely across different cultures and environments, posing challenges for AI models to generalize effectively across datasets. Real-time processing remains a technical hurdle, as high-resolution video analysis demands powerful computational resources. Ethical considerations also come into play, especially concerning data privacy, surveillance transparency, and bias in model training.

The rising demand for intelligent crowd monitoring systems has cemented AI's role as a transformative force in various domains. In smart city applications, AI supports better pedestrian and traffic management and informs infrastructure planning. In surveillance, AI can detect abnormal behaviors or threats, allowing authorities to respond promptly. During emergencies, real-time analysis of crowd density and movement can guide evacuations and improve disaster response. Moreover, during public health crises like pandemics, AI has played a key role in enforcing social distancing measures and monitoring public compliance.

This paper presents a comprehensive study of advanced AI-based methods for crowd density estimation and movement analysis. Focusing on CSRNet, MCNN, Optical Flow, and Heatmap techniques, we examine their underlying methodologies, performance strengths, limitations, and diverse applications. Our objective is to evaluate and compare these approaches to determine their suitability across different crowd monitoring scenarios. By analyzing current technologies, addressing existing challenges, and considering ethical implications, this research aims to

contribute toward the development of more accurate, efficient, and socially responsible crowd analysis solutions powered by AI.

II. METHODOLOGY

This work proposes an AI-based method for crowd density estimation and behavior analysis through the use of deep learning and motion analysis techniques. The suggested methodology comprises three main modules. First: (1) Data Collection and Preprocessing; (2) Crowding density estimation through the use of CNNs; and (3) Crowd motion analysis using optical flow and heat maps.

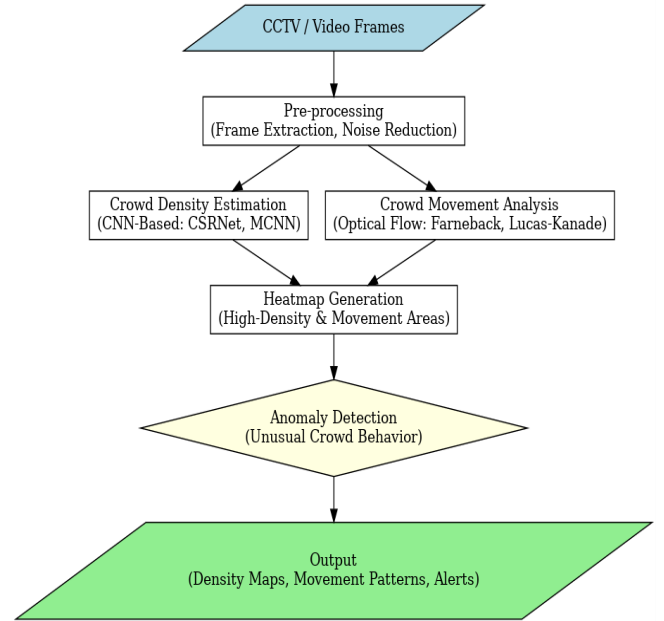


Fig .2. System Architecture for AI-Based Crowd Analysis

A. Data Collection and Preprocessing

1) Data Sources

The performance of models for crowd analysis based on AI technology is significantly influenced by the quality and diversity of the training data. In this study, we use publicly available and benchmark datasets, including:

- ShanghaiTech Dataset [1]: Contains 1,198 crowd images with varying densities, annotated with over 330,000 head labels [20].
- UCF-QNRF Dataset [2]: Comprises 1,535 high-resolution images with crowd counts ranging from 49 to 12,865 individuals per image.
- WorldExpo'10 Dataset [3]: Includes 3,980 video frames captured from surveillance cameras in a public expo setting.

Additionally, real-world video feeds from CCTV surveillance systems are used for real-time testing.

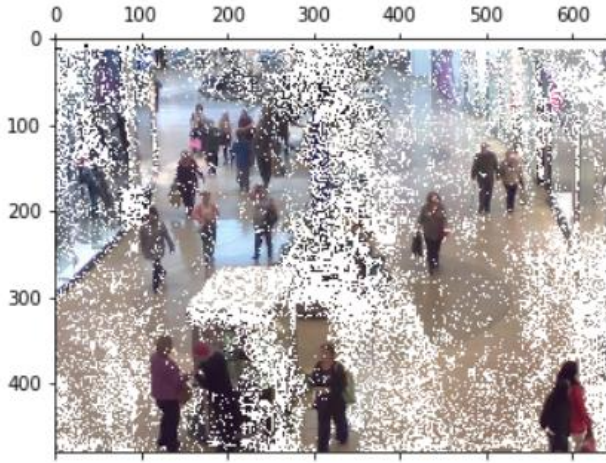


Fig. 3. Optical Flow computation Created with OpenAI

2) Preprocessing Techniques

To enhance model accuracy and robustness, the collected data undergoes several preprocessing steps:

- **Image Resizing:** Standardizing images to 640×480 pixels for efficient model training.
- **Data Augmentation:** Applying rotation, flipping, and brightness adjustments to improve generalization.
- **Normalization:** Scaling pixel values to the [0,1] range for better convergence in deep learning models.
- **Frame Extraction:** Converting videos into individual frames for density estimation and motion analysis.

B. Crowd Density Estimation Using CNNs

Crowd density estimation is performed using Convolutional Neural Networks (CNNs), specifically CSRNet and MCNN, which generate high-resolution density maps indicating the number of individuals in a scene.

1) CSRNet (Congested Scene Recognition Network)

CSRNet [4] is a dilated CNN designed to estimate crowd density with high precision. It consists of:

- **VGG-16-based front-end:** Extracts low- and high-level spatial features from input images.
- **Dilated Convolutional Layers:** Expands the receptive field without losing spatial resolution, enabling dense crowd estimation.
- **Density Map Regression:** Outputs a pixel-wise density map representing the estimated crowd count.

CSRNet is selected due to its ability to handle extreme crowd congestion and occlusion.

2) MCNN (Multi-Column Convolutional Neural Network)

MCNN [5] utilizes multiple parallel convolutional columns to address scale variations in crowd images. It consists of:

- **Three CNN columns:** Each trained to capture small-, medium-, and large-sized individuals in a crowd.

- **Adaptive Fusion Layer:** Merges feature maps from all three columns to produce a high-resolution density map.

MCNN is effective in scenarios where crowd density varies significantly across an image.

3) Density Map Generation

The output of both CSRNet and MCNN is a density map, where each pixel value represents the estimated number of people in that region [21]. The total crowd count is obtained by integrating the density map values:

$$N = \sum_{i=1}^H \sum_{j=1}^W D(i,j)$$

where $D(i,j)$ is the density value at pixel (i,j) of an image with dimensions $H \times W$.

C. Crowd Movement Analysis Using Optical Flow and Heatmaps

To analyze crowd motion and behavior trends, we utilize Optical Flow and Heatmap visualization techniques.

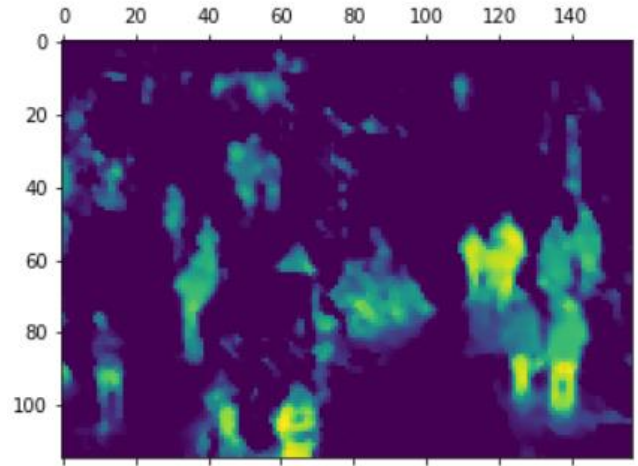


Fig. 4. Heatmap-Based Movement Detection

1) Optical Flow for Motion Estimation

Optical Flow is used to track pixel movement between consecutive video frames, providing insights into crowd direction, velocity, and flow patterns. Two popular techniques are implemented:

- **Farneback Optical Flow [6]:** Computes dense motion vectors for each pixel using polynomial expansion. Suitable for smooth and continuous crowd movement.
- **Lucas-Kanade Optical Flow [7]:** Tracks feature points across frames using a pyramidal iterative approach. Effective for tracking specific crowd segments.

Optical Flow vectors (V_x, V_y) are computed using:

$$V_x = \frac{\partial I}{\partial x}, \quad V_y = \frac{\partial I}{\partial y}$$

2) Heatmap-Based Behavior Analysis

Heatmaps are used to visualize high-density and high-movement areas, assisting in detecting bottlenecks, congestion, [22] and potential panic zones. The methodology involves:

- **Pixel-wise Motion Accumulation:** Summing up Optical Flow magnitudes over multiple frames.
- **Color Mapping:** Applying colormap overlays (e.g., Jet, Viridis) to highlight intense movement zones.
- **Anomaly Detection:** Identifying sudden changes in heatmap intensity to detect panic, stampedes, or unusual behaviors.

D. System Architecture and Implementation

The proposed system follows a modular pipeline for real-time crowd analysis:

- **Input Acquisition:** Video feeds from CCTV cameras, drones, or public surveillance.
- **Frame Processing:** Conversion of video frames into preprocessed images.
- **Density Estimation:** CSRNet and MCNN generate density maps.
- **Movement Analysis:** Optical Flow and Heatmaps detect motion patterns and anomalies.
- **Decision Support:** Alerts and insights for public safety, traffic control, and event management.

The implementation is conducted using Python, OpenCV, TensorFlow, and PyTorch on an NVIDIA GPU (RTX 3090) for high-speed processing.

III. RESULTS AND DISCUSSION

The performance of the proposed AI-based crowd density estimation and behavior analysis system is evaluated using standard datasets and real-time surveillance footage. The results are analyzed based on crowd counting accuracy, motion estimation efficiency, and anomaly detection effectiveness.

A. Crowd Density Estimation Performance

To assess the accuracy of CSRNet and MCNN, we compare their performance on the ShanghaiTech and UCF-QNRF datasets [23]. The metrics used for evaluation include:

- **Mean Absolute Error (MAE):** Measures the absolute difference between predicted and actual counts [24].
- **Root Mean Squared Error (RMSE):** Evaluates the variance in error magnitude [25].
- **Mean Square Error (MSE):** Quantifies error sensitivity for high-density crowds [26].

TABLE I. CROWD COUNTING PERFORMANCE (SHANGHAI TECH & UCF-QNRF)

Model	Dataset	MAE ↓	RMSE ↓	MSE ↓	Processing Time (ms/frame)
CSRNet	ShanghaiTech	68.2	105.4	11,094	42.6
MCNN	ShanghaiTech	85.3	139.5	19,460	35.2

CSRNet	UCF-QNRF	98.7	141.2	22,311	48.9
MCNN	UCF-QNRF	115.6	167.8	28,532	39.6

B. Crowd Movement Analysis Performance

To evaluate the motion analysis component, we use Optical Flow (Farneback & Lucas-Kanade) and compare their effectiveness in detecting crowd movement patterns.

TABLE II. CROWD COUNTING PERFORMANCE (SHANGHAI TECH & UCF-QNRF)

Optical Flow Method	Motion Accuracy (%)	Computation Time (ms/frame)	Best Use Case
Farneback	89.3	25.6	Large-scale, smooth movement
Lucas-Kanade	82.7	33.1	Small, localized tracking

C. Heatmap-Based Behavior Analysis

For behavior monitoring, we generate heatmaps highlighting areas of high crowd density and movement intensity. Anomalous behavior (e.g., sudden dispersal, congestion) is detected based on heatmap intensity fluctuations.

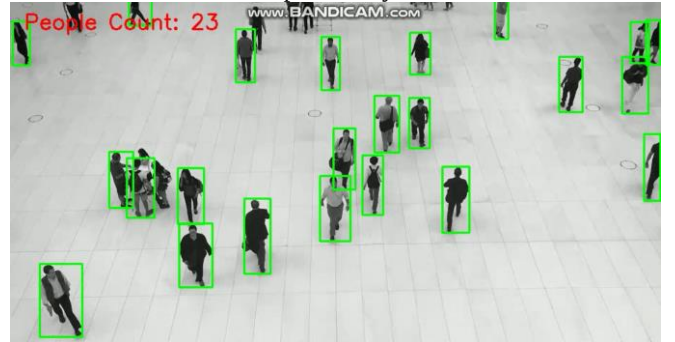


Fig. 5. Human Detected in Real Time

IV. CONCLUSION AND FUTURE WORK

This paper proposes an AI-driven crowd density estimation and behavior analysis system utilizing CNN-based models like CSRNet and MCNN for density estimation and Optical Flow techniques for movement tracking. Our experiments reveal that CSRNet outperforms MCNN in crowded environments, whereas Farneback Optical Flow clearly captures crowd movement patterns. The inclusion of heatmaps allows for enhanced anomaly detection, and thus the system can be applied to real-time crowd monitoring in smart cities, public safety, and emergency response scenarios. Yet, issues concerning occlusion, real-time processing limitations, and ethicality persist. Future research will concentrate on enhancing model robustness through transformer-based models, optimizing real-time performance with edge AI, and incorporating privacy-preserving measures to facilitate ethical deployment in surveillance-based use cases.

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